

JEE Mains 2019 Chapter wise Question Bank

Hyperbola – Questions

Q1

Let $0 < \theta < \frac{\pi}{2}$. If the eccentricity of the hyperbola

$\frac{x^2}{\cos^2 \theta} - \frac{y^2}{\sin^2 \theta} = 1$ is greater than 2, then the length of

its latus rectum lies in the interval:

- (1) $(3, \infty)$ (2) $(3/2, 2]$
(3) $(2, 3]$ (4) $(1, 3/2]$

9 Jan Morning

Q2

A hyperbola has its centre at the origin, passes through the point (4, 2) and has transverse axis of length 4 along the x -axis. Then the eccentricity of the hyperbola is :

- (1) $\frac{3}{2}$ (2) $\sqrt{3}$
(3) 2 (4) $\frac{2}{\sqrt{3}}$

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Q3

The equation of a tangent to the hyperbola $4x^2 - 5y^2 = 20$ parallel to the line $x - y = 2$ is:

- (1) $x - y + 1 = 0$ (2) $x - y + 7 = 0$
(3) $x - y + 9 = 0$ (4) $x - y - 3 = 0$

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Q4

Let

$$S = \left\{ (x, y) \in \mathbf{R}^2 : \frac{y^2}{1+r} - \frac{x^2}{1-r} = 1 \right\},$$

where $r \neq \pm 1$ Then S represents:

- (1) a hyperbola whose eccentricity is

$$\frac{2}{\sqrt{1-r}}, \text{ when } 0 < r < 1$$

- (2) an ellipse whose eccentricity is

$$\sqrt{\frac{2}{r+1}}, \text{ when } r > 1$$

- (3) a hyperbola whose eccentricity is

$$\frac{2}{\sqrt{r+1}}, \text{ when } 0 < r < 1$$

- (4) an ellipse whose eccentricity is

$$\frac{1}{\sqrt{r+1}}, \text{ when } r > 1$$

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Q5

Equation of a common tangent to the parabola $y^2 = 4x$ and the hyperbola $xy = 2$ is :

- (1) $x + y + 1 = 0$ (2) $x - 2y + 4 = 0$
(3) $x + 2y + 4 = 0$ (4) $4x + 2y + 1 = 0$

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Q6

If a hyperbola has length of its conjugate axis equal to 5 and the distance between its foci is 13, then the eccentricity of the hyperbola is :

- (1) $\frac{13}{12}$ (2) 2 (3) $\frac{13}{6}$ (4) $\frac{13}{8}$

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Hyperbola

Q7

If the vertices of a hyperbola be at $(-2, 0)$ and $(2, 0)$ and one of its foci be at $(-3, 0)$, then which one of the following points does not lie on this hyperbola?

- (1) $(-6, 2\sqrt{10})$ (2) $(2\sqrt{6}, 5)$
(3) $(4, \sqrt{15})$ (4) $(6, 5\sqrt{2})$

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Q8

If the eccentricity of the standard hyperbola passing through the point $(4, 6)$ is 2, then the equation of the tangent to the hyperbola at $(4, 6)$ is :

- (1) $x - 2y + 8 = 0$ (2) $2x - 3y + 10 = 0$
(3) $2x - y - 2 = 0$ (4) $3x - 2y = 0$

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Q9

If the line $y = mx + 7\sqrt{3}$ is normal to the hyperbola

$\frac{x^2}{24} - \frac{y^2}{18} = 1$, then a value of m is :

- (1) $\frac{\sqrt{5}}{2}$ (2) $\frac{\sqrt{15}}{2}$
(3) $\frac{2}{\sqrt{5}}$ (4) $\frac{3}{\sqrt{5}}$

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Q10

If a directrix of a hyperbola centred at the origin and passing through the point $(4, -2\sqrt{3})$ is $5x = 4\sqrt{5}$ and its eccentricity is e , then :

- (1) $4e^4 - 24e^2 + 27 = 0$ (2) $4e^4 - 12e^2 - 27 = 0$
(3) $4e^4 - 24e^2 + 35 = 0$ (4) $4e^4 + 8e^2 - 35 = 0$

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Q11

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If $5x + 9 = 0$ is the directrix of the hyperbola $16x^2 - 9y^2 = 144$, then its corresponding focus is :

- (1) $(5, 0)$ (2) $\left(-\frac{5}{3}, 0\right)$
(3) $\left(\frac{5}{3}, 0\right)$ (4) $(-5, 0)$

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Hyperbola - Solutions

Q1

$$(1) \because a^2 = \cos^2 \theta, b^2 = \sin^2 \theta$$

$$\text{and } e > 2 \Rightarrow e^2 > 4 \Rightarrow 1 + b^2/a^2 > 4$$

$$\Rightarrow 1 + \tan^2 \theta > 4$$

$$\Rightarrow \sec^2 \theta > 4 \Rightarrow \theta \in \left(\frac{\pi}{3}, \frac{\pi}{2} \right)$$

Latus rectum,

$$LR = \frac{2b^2}{a} = \frac{2\sin^2 \theta}{\cos \theta} = 2(\sec \theta - \cos \theta)$$

$$\Rightarrow \frac{d(LR)}{d\theta} = 2(\sec \theta \tan \theta + \sin \theta) > 0 \forall \theta \in \left(\frac{\pi}{3}, \frac{\pi}{2} \right)$$

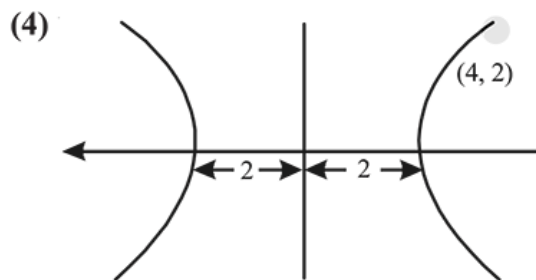
$$\therefore \min(LR) = 2\left(\sec \frac{\pi}{3} - \cos \frac{\pi}{3}\right) = 2\left(2 - \frac{1}{2}\right) = 3$$

$$\max(LR) \text{ tends to infinity as } \theta \rightarrow \frac{\pi}{2}$$

Hence, length of latus rectum lies in the interval $(3, \infty)$

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Q2



Consider equation of hyperbola

$$\frac{x^2}{2^2} - \frac{y^2}{b^2} = 1$$

 $\therefore (4, 2)$ lies on hyperbola

$$\therefore \frac{16}{4} - \frac{4}{b^2} = 1$$

$$\therefore b^2 = \frac{4}{3}$$

$$\text{Since, eccentricity} = \sqrt{1 + \frac{b^2}{a^2}}$$

$$\text{Hence, eccentricity} = \sqrt{1 + \frac{\frac{4}{3}}{4}} = \sqrt{1 + \frac{1}{3}} = \frac{2}{\sqrt{3}}$$

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Q3

Hyperbola

(1) Given, the equation of line,

$$x - y = 2 \Rightarrow y = x - 2$$

$\therefore$ its slope = $m = 1$

Equation of hyperbola is:

$$\frac{x^2}{5} - \frac{y^2}{4} = 1$$

$$\Rightarrow a^2 = 5, b^2 = 4$$

The equation of tangent to the hyperbola is,

$$y = mx \pm \sqrt{a^2 m^2 - b^2}$$
$$= x \pm \sqrt{5 - 4}$$

$$\Rightarrow y = x \pm 1$$

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Q4

(2) Since, $r \neq \pm 1$, then there are two cases, when $r > 1$

$$\frac{x^2}{r-1} + \frac{y^2}{r+1} = 1 \quad (\text{Ellipse})$$

Then,

$$(r-1) = (r+1)(1-e^2) \Rightarrow 1-e^2 = \frac{(r-1)}{(r+1)}$$

$$\Rightarrow e^2 = 1 - \frac{(r-1)}{(r+1)} = \frac{2}{(r+1)}$$

$$\Rightarrow e = \sqrt{\frac{2}{(r+1)}}$$

When $0 < r < 1$, then

$$\frac{x^2}{1-r} - \frac{y^2}{1+r} = -1 \quad (\text{Hyperbola})$$

Then,

$$(1-r) = (1+r)(e^2-1) \Rightarrow e^2 = 1 + \frac{(r-1)}{(r+1)} = \frac{2r}{(r+1)}$$

$$\Rightarrow e = \sqrt{\frac{2r}{r+1}}$$

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Q5

(3) Equation of a tangent to parabola $y^2 = 4x$ is:

$$y = mx + \frac{1}{m}$$

This line is a tangent to $xy = 2$

$$\therefore x\left(mx + \frac{1}{m}\right) = 2$$

$$\Rightarrow mx^2 + \frac{1}{m}x - 2 = 0$$

$\therefore$ Tangent is common for parabola and hyperbola.

$$\therefore D = \left(\frac{1}{m}\right)^2 - 4 \cdot m \cdot (-2) = 0$$

$$\frac{1}{m^2} + 8m = 0$$

$$1 + 8m^3 = 0$$

$$m^3 = -\frac{1}{8}$$

$$m = -\frac{1}{2}$$

$$\therefore \text{Equation of common tangent: } y = -\frac{1}{2}x - 2$$

$$\Rightarrow 2y = -x - 4$$

$$\Rightarrow x + 2y + 4 = 0$$

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Q6

Hyperbola

(1) $\therefore$ Conjugate axis = 5

$$\therefore 2b = 5$$

Distance between foci = 13

$$2ae = 13$$

$$\text{Then, } b^2 = a^2(e^2 - 1)$$

$$\Rightarrow a^2 = 36$$

$$\therefore a = 6$$

$$ae = \frac{13}{2} \Rightarrow e = \frac{13}{12}$$

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Q7

(4) Let the points are,

$$A(2, 0), A'(-2, 0) \text{ and } S(-3, 0)$$

$\Rightarrow$ Centre of hyperbola is $O(0, 0)$

$$AA' = 2a \Rightarrow 4 = 2a \Rightarrow a = 2$$

$\therefore$ Distance between the centre and foci is ae .

$$\therefore OS = ae \Rightarrow 3 = 2e \Rightarrow e = \frac{3}{2}$$

$$\Rightarrow b^2 = a^2(e^2 - 1) = a^2e^2 - a^2 = 9 - 4 = 5$$

$$\Rightarrow \text{Equation of hyperbola is } \frac{x^2}{4} - \frac{y^2}{5} = 1 \quad \dots(1)$$

$\therefore$ (6, 552) does not satisfy eq (1).

$\therefore$ (6, $5\sqrt{2}$) does not lie on this hyperbola.

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Q8

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(3) Let equation of hyperbola be

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \dots(i)$$

$$\therefore e = \sqrt{1 + \frac{b^2}{a^2}} \Rightarrow b^2 = a^2(e^2 - 1)$$

$$e = 2 \Rightarrow b^2 = 3a^2 \quad \dots(ii)$$

Equation (i) passes through (4, 6),

$$\therefore \frac{16}{a^2} - \frac{36}{b^2} = 1 \quad \dots(iii)$$

On solving (i) and (ii), we get

$$a^2 = 4, b^2 = 12$$

$$\text{Now equation of hyperbola is } \frac{x^2}{4} - \frac{y^2}{12} = 1$$

Now equation of tangent to the hyperbola at (4, 6) is

$$\frac{4x}{4} - \frac{6y}{12} = 1 \Rightarrow x - \frac{y}{2} = 1 \Rightarrow 2x - y = 2$$

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Q9

(3) Since, $lx + my + n = 0$ is a normal to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$,

$$\text{then } \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$$

but it is given that $mx - y + 7\sqrt{3}$ is normal to hyperbola

$$\frac{x^2}{24} - \frac{y^2}{18} = 1$$

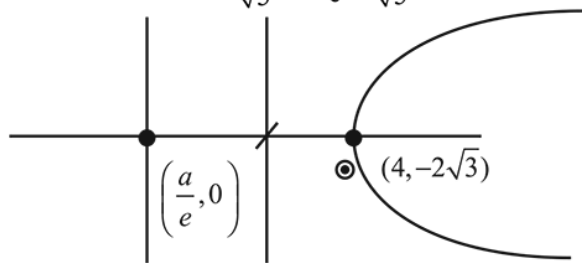
$$\text{then } \frac{24}{m^2} - \frac{18}{(-1)^2} = \frac{(24+18)^2}{(7\sqrt{3})^2} \Rightarrow m = \frac{2}{\sqrt{5}}$$

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Q10

(3) $\therefore$ directrix of a hyperbola is,

$$5x = 4\sqrt{5} \Rightarrow x = \frac{4}{\sqrt{5}} \Rightarrow \frac{a}{e} = \frac{4}{\sqrt{5}}$$



Now, hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through $(4, -2\sqrt{3})$

$$\therefore \frac{16}{a^2} - \frac{12}{a^2 e^2 - a^2} = 1$$

$$\left[\because e^2 = 1 + \frac{b^2}{a^2} \Rightarrow a^2 e^2 - a^2 = b^2 \right]$$

$$\Rightarrow \frac{4}{a^2} \left[\frac{4}{1} - \frac{3}{e^2 - 1} \right] = 1 \Rightarrow 4e^2 - 4 - 3 = (e^2 - 1) \left(\frac{a^2}{4} \right)$$

$$\Rightarrow 4(4e^2 - 7) = (e^2 - 1) \left(\frac{4e}{\sqrt{5}} \right)^2$$

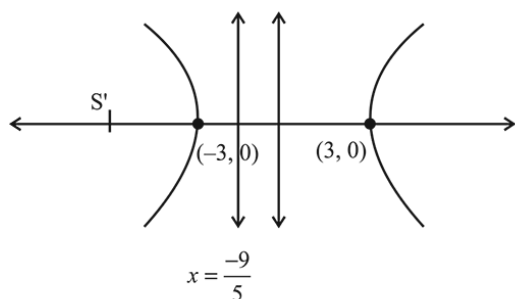
$$\Rightarrow 4e^4 - 24e^2 + 35 = 0$$

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Q11

$$(4) 16x^2 - 9y^2 = 144 \Rightarrow \frac{x^2}{9} - \frac{y^2}{16} = 1$$

Then focus is $S'(-ae, 0)$



$$a = 3, b = 4 \Rightarrow e^2 = 1 + \frac{16}{9} = \frac{25}{9} \quad \left[\because e = \sqrt{1 + \frac{b^2}{a^2}} \right]$$

$$\therefore \text{the focus } S' \equiv \left(3 - \frac{5}{3}, 0 \right) \equiv (-5, 0)$$

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